

EXECUTIVE SUMMARY

ABC Insurance CO is a regional auto insurer and wants to expand their market share while maintaining its current customer base. Several potential expansion states in the US require car insurance policy rates to be gender neutral and ABC needs to evaluate the status of its current offerings. Additionally, they want to make sure that they aren't offering newer customers better rates than long-time customers – which puts them at risk of some losing current customers to competitors. They also want to make sure that they have adequate staffing levels to handle increased demand throughout the year in order to retain their existing customers. Finally, to best plan their marketing spend, ABC has collected demographic data about past sales leads that did or did not sign-up for insurance. They'd like to use this data to predict which current sales leads are most likely to buy a new policy from them.

Major Finding(s)

- Gender significantly affects premium with females paying \$42.07 higher annually. *(Appendix 2)*
- Vintage significantly affects annual premiums. Customers with higher vintage pay more (about \$0.90 per day since enrolled). *(Appendix 2)*.
- While YOY increase is projected to be nearly 20%, sales are much higher in the last 4 months of the year, which account for over 47% of sales. *(Appendix 4)*
- Past performance analysis shows key indicators can target a higher rate of successful policy conversions. *(Appendix 3)*

Recommendation(s) for Action

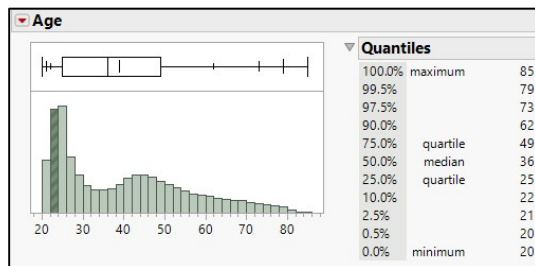
1. **Address inconsistencies in premium rates by gender.** Adjust premiums downward for females above the level predicted by the model, focusing on those furthest from path.
2. **Review premiums incentives for customer longevity.** Identify promotional incentive programs for existing customers to retain loyalty.
3. **Ongoing Monitoring.** Predict and prevent unequal premium amounts based on gender and customer length.
4. **Proactive temporary hiring.** Increase staffing by 20% from start of Q3 to Q4 end. *(Appendix 4)*
5. **Targeted Expansion of sales pipeline and marketing.** Focus premium conversions on previously insured customers, without vehicle damage, and vehicles less than 1 year old.

Analytical Overview

A comprehensive assessment of the data was conducted to determine distributions and summaries of the sample data, and other statistics from historical customer demographics. Multiple linear regression modeling was conducted to determine which factors most impact rates and identify necessary adjustments needed. A logistic regression model was developed to determine which leads were most likely to convert based on past sales data. Past sales data was analyzed over time and employed to forecast future improvements.

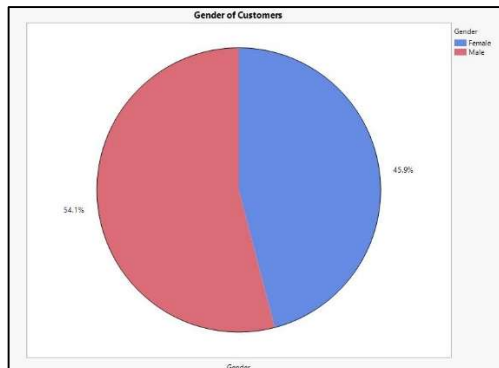
Appendix 1: Exploratory Data Analysis

Customer analysis: The average age of X Insurance's customers is 39, with 50% of policyholders being between the ages of 25 and 49, and a significant portion of customers are around 24 years old. The average Vintage (number of days a customer has been associated with the company) is 154.35 days, with a minimum of 10 days and a maximum of 299 days.

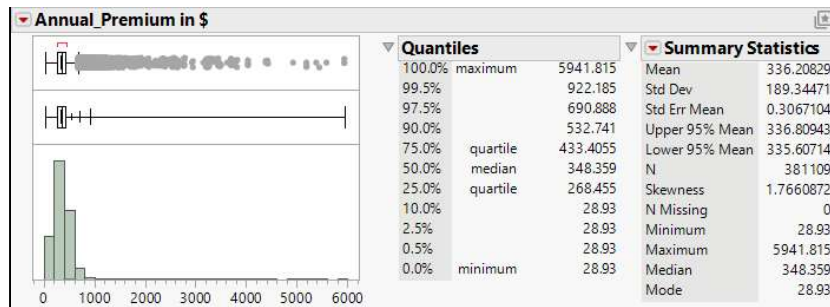


	Mean	Median	Mode	Min	Max
Age	38.822583565	36	24	20	85
Vintage	154.34739668	154	256	10	299

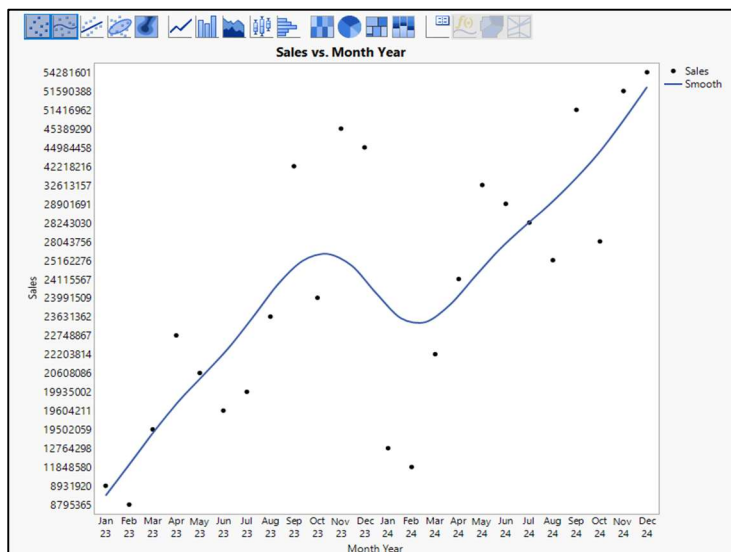
Roughly 54% of customers are Male and 46% are female.



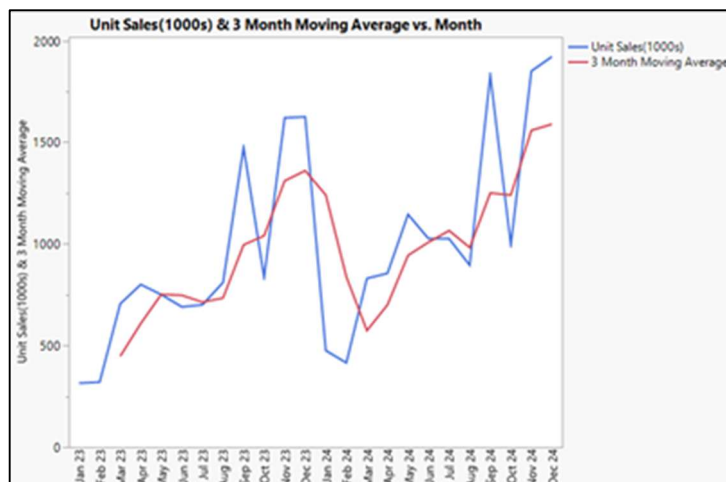
Premium analysis: While half of all policyholder's Annual Premium lies between \$268.46 and \$433.41 per year, overall customer premiums vary wildly (the highest reaching up to \$5,942).



Sales analysis: X Insurance Co has provided car sales data in their expansion region as a proxy for increased demand for auto insurance policies. Based on our initial analysis of sales from January 2023 – December 2024, we can see an upward trend YOY, with a slight lapse from November 2023 – February 2024 suggesting seasonality. With this data, we expect to be able to build a multilinear regression model to predict new client market demand for auto insurance in future years.



Time Series Car Sales with 3 Month Moving Average:



SEASON PATTERN

JAN-FEB	Lowest annual sales
MAR-MAY	Moderate increase (Spring bump)
JUN-AUG	Slower summer months
SEP	Very large spike
NOV-DEC	Highest of the year

Appendix 2: Determining Effects of Gender and Vintage with Multiple Linear Regression

Model Selection - Preprocessing

Columns Considered

- ID – record ID
- Annual Premium – continuous target you can use for multiple linear regression
- Age – customer age
- Gender – Male/Female
- Previously Insured – 0 = no, 1 = yes
- Vehicle Age – < 1 Year, 1-2 Year, > 2 Years
- Vehicle Damage – Yes/No
- Region Code – integer region ID
- Policy Sales Channel – channel ID
- Vintage – days since first contact

Dummy variables were created for the following:

- Gender (binary)
- Previously Insured (binary)
- Vehicle Age (2 variables)
- Vehicle damage (binary)

The data included information about each policy holder's region (1-40) and sales channels (1-30). Without additional information, it was necessary to treat this as categorical data which required an additional 68 dummy variables to accommodate the data which impacted adjusted R² values and overall model reliability. Using hypothesis testing we found that only 12 of 40 dummy variables for region and 15 of 30 for Channel contributed significantly to the overall model.

Without additional information that allowed us to “bucket” the data or somehow transform it from direct categories, we decided to remove these variables from the analysis. Preliminary modeling shows that the removal of these variables resulted in losing about 1% of the explained variability in the model.

Summary of Fit	
RSquare	0.849193
RSquare Adj	0.84631
Root Mean Square Error	49.118
Mean of Response	462.1035
Observations (or Sum Wgts)	4000

Model Selection

MLR Equation:

$$\begin{aligned}
 \text{Predicted Premium} = & 175.76 + 3.13(\text{Age}) + 42.07(\text{Female}) \\
 & + 39.52(\text{Not Previously Insured}) - 22.15(\text{Vehicle} < 1 \text{ Year}) \\
 & + 26.20(\text{Vehicle} > 2 \text{ Years}) - 37.78(\text{No Vehicle Damage}) \\
 & + 0.90(\text{Vintage})
 \end{aligned}$$

Model	Number	RSquare	RMSE	AICc	BIC	Cp
Vintage	1	0.3593857	100.2931	48220.29	48239.16	11602.98
Age,Vintage	2	0.5131786	87.44032	47124.17	47149.34	7860.12
Age,Gender{Male-Female},Vintage	3	0.6245632	76.79796	46086.95	46118.4	5149.9
Age,Gender{Male-Female},Previously_Insured{1-0},Vintage	4	0.722411	66.04452	44881.16	44918.9	2769.304

Age,Gender{Male-Female},Previously_Insured{1-0},Vehicle_Damage{No-Yes},Vintage	5	0.8137452	54.10579	43287.06	43331.09	547.3128
Age,Gender{Male-Female},Previously_Insured{1-0},Vehicle_Age{< 1 Year&1-2 Year-> 2 Years},Vehicle_Damage{No-Yes},Vintage	6	0.8324865	51.31795	42864.86	42915.18	92.96091
Age,Gender{Male-Female},Previously_Insured{1-0},Vehicle_Age{< 1 Year&1-2 Year-> 2 Years},Vehicle_Age{< 1 Year-1-2 Year},Vehicle_Damage{No-Yes},Vintage	7	0.8360578	50.77433	42780.67	42837.27	8

The model shows an adjusted R^2 of .836, meaning that it explains 83% of the variability in the price of the annual premium. Although the R value is not above 90%, it still contributes substantially to predicting the insurance premiums. Additionally, the lack of fit test shows good overall fit in the model and the low VIF scores show low probability of multicollinearity in the independent variables.

Summary of Fit

RSquare				0.836058		
RSquare Adj				0.83577		
Root Mean Square Error				50.77433		
Mean of Response				462.1035		
Observations (or Sum Wgts)				4000		

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio		
Model	7	52483681	7497669	2908.291		
Error	3992	10291505	2578		Prob > F	
C. Total	3999	62775186			<.0001*	

Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio		
Lack Of Fit	3972	10224450	2574.13	0.7678		
Pure Error	20	67054	3352.71		Prob > F	
Total Error	3992	10291505			0.8349	
				Max RSq	0.9989	

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t	VIF	
Intercept	175.76046	3.001031	58.57	<.0001*		
Age	3.1263987	0.054698	57.16	<.0001*	1.0030355	
Gender[Female]	42.067718	0.803177	52.38	<.0001*	1.0009025	
Previously_Insured[0]	39.521358	0.818541	48.28	<.0001*	1.0014444	
Vehicle_Age[< 1 Year]	-22.14555	1.183233	-18.72	<.0001*	1.5027965	
Vehicle_Age[> 2 Years]	26.202356	1.182595	22.16	<.0001*	1.5045588	
Vehicle_Damage[No]	-37.78459	0.80317	-47.04	<.0001*	1.0007205	
Vintage	0.8978698	0.009589	93.64	<.0001*	1.00022	

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F	
Age	1	1	8422280	3266.941	<.0001*	
Gender	1	1	7072345	2743.311	<.0001*	
Previously_Insured	1	1	6009951	2331.216	<.0001*	
Vehicle_Age	2	2	1400678	271.6564	<.0001*	
Vehicle_Damage	1	1	5705626	2213.171	<.0001*	
Vintage	1	1	22604996	8768.314	<.0001*	

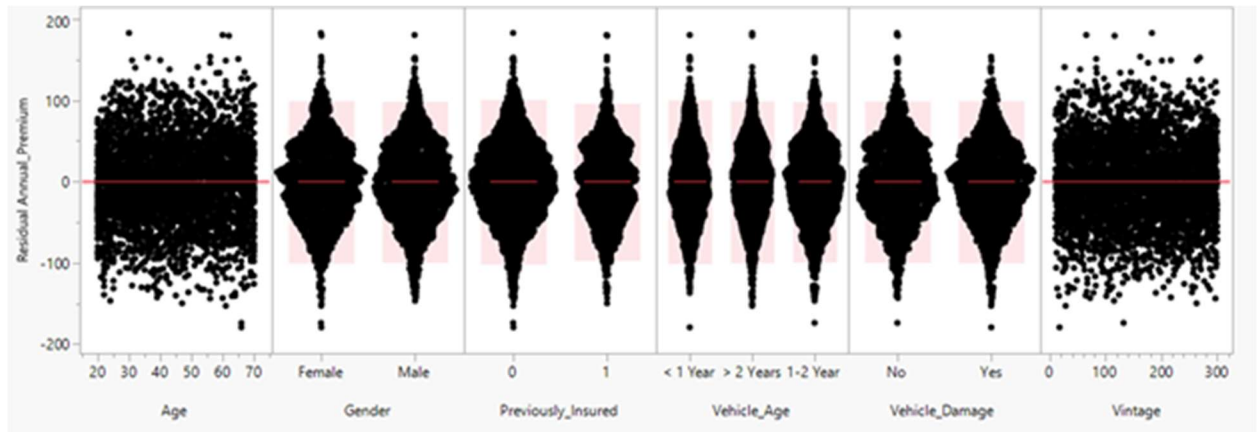
Effect Details

Durbin-Watson

Durbin-Watson	Number of Obs.	AutoCorrelation	Prob<DW			
1.9775025	4000	0.0110	0.2378			

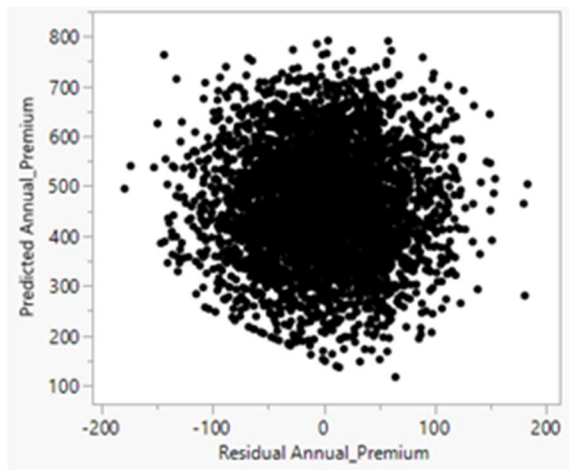
Model Validation – Residual Analysis

1. Linearity



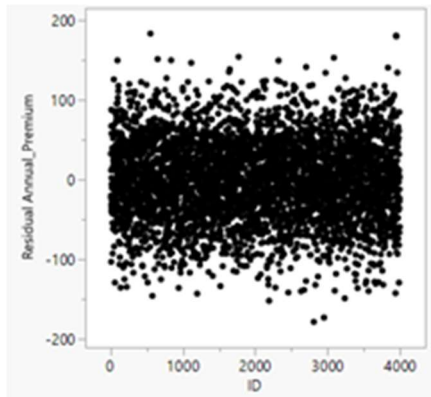
Residuals vs independent variable values scatter randomly around 0 for continuous and evenly above and below zero for each categorical value shows that the relationship between the dependent and independent variables is linear and additive.

2. Equal Variance



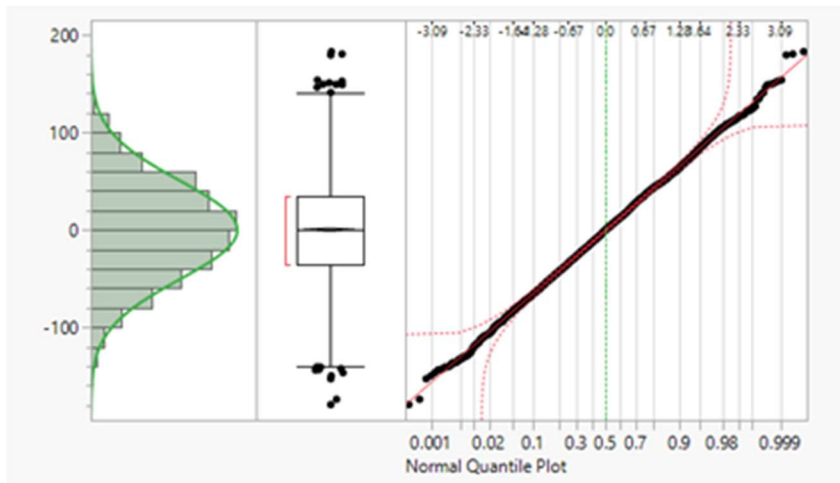
The scatter plot of residual vs predicated values and the Durbin Watson score near 2.0 show that the variables are homoscedastic to each other.

3. Independence of Errors



Even spread of data indicates residuals are not correlated with each other.

4. Normality of the Residuals



The residuals show a normal distribution with linear probability plot

5. Cross Validation

Crossvalidation			
Source	RSquare	RASE	Freq
Training Set	0.8404	49.947	2787
Validation Set	0.8254	52.527	1213

The training set and the validation set have similar R^2 values, indicating the model with predict new data at a similar level of accuracy.

Conclusion: Conditions for using a Multiple Linear Regression model have been met and MLR is a suitable model for the given data.

Hypothesis Testing

1. Hypothesis Test for Gender

Hypotheses

$$H_0: \beta_{\text{Gender}} = 0 \text{ (Gender has no effect on premium)}$$

$$H_a: \beta_{\text{Gender}} \neq 0 \text{ (Gender affects premium)}$$

Because:

- $p < 0.05$, and
- t is extremely large ($|t| = 52.38$),

we reject the null hypothesis.

Conclusion: Gender significantly affects Annual Premium. This means females (coded = 1) pay \$42.07 more, on average, holding all else constant.

2. Hypothesis Test for Vintage

Hypotheses

$$H_0: \beta_{\text{Vintage}} = 0 \text{ (Vintage has no effect on premium)}$$

$$H_a: \beta_{\text{Vintage}} \neq 0 \text{ (Vintage affects premium)}$$

Because:

- $p < 0.05$, and
- The t -value is extremely large (93.64),

We reject the null hypothesis.

Conclusion: Vintage significantly affects Annual Premium. Customers with higher Vintage pay more (about \$0.90 per day since enrolled).

Appendix 3: Estimating which Leads Will Buy New Auto Insurance with Binomial Logistical Regression Modeling

Develop Model Based on Past Leads – All Variables

Parameter Estimates						
Term	Estimate	Std Error	L-R		Lower CL	Upper CL
Intercept	1.382894	0.2700791	26.21778	<.0001*	0.853439	1.912349
Gender[Female]	0.0611766	0.0455801	1.8014404	0.1795	-0.028177	0.1505306
Age	0.025674	0.0043465	34.89123	<.0001*	0.0171533	0.0341946
Vehicle_Age[< 1 Year]	1.0829809	0.1148299	88.94701	<.0001*	0.8578717	1.3080901
Vehicle_Age[> 2 Years]	-0.678419	0.1133298	35.835043	<.0001*	-0.900588	-0.456251
Vehicle_Damage[No]	1.2057611	0.1625878	54.997992	<.0001*	0.8870289	1.5244933
Annual Premium	-0.000153	0.0003445	0.196964	0.6572	-0.000828	0.0005225
Previously_Insured	3.3904983	0.6106168	30.831134	<.0001*	2.1934632	4.5875334

Which predictors are statistically significant?

Predictor	Chi-Square	p-value	Significant?
Gender	1.80	0.1786	✗ Not significant
Age	34.89	< .0001	✓ Significant
Vehicle Age	96.06	< .0001	✓✓ Highly significant
Vehicle Damage	54.98	< .0001	✓✓ Highly significant
Annual Premium	0.20	0.6585	✗ Not significant
Previously Insured	30.83	< .0001	✓ Significant

Model Equation: (Gender & Annual Premium removed)

$$\log\left(\frac{p}{1-p}\right) = 1.3288917$$

$$+ 0.0253834(\text{Age})$$

$$+ 1.0946418(\text{VehicleAge} < 1 \text{ Year})$$

$$- 0.6866011(\text{VehicleAge} > 2 \text{ Years})$$

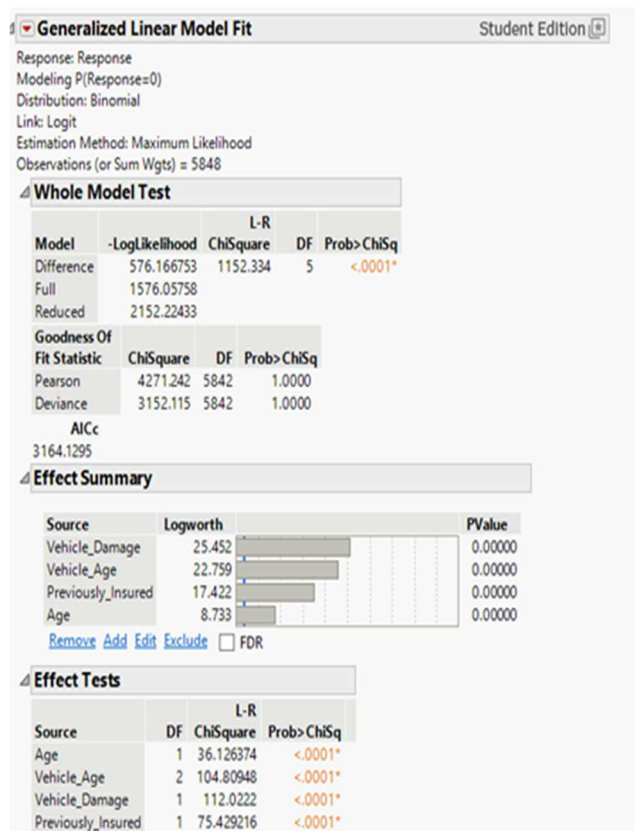
$$+ 1.2089982(\text{VehicleDamage} = \text{No})$$

$$+ 3.3839828(\text{Previously Insured})$$

Parameter Estimates				
Term	Estimate	Std Error	L-R	Prob>ChiSq
Intercept	1.3288917	0.2454635	29.309297	<.0001*
Age	0.0253834	0.0043156	34.595065	<.0001*
Vehicle_Age[< 1 Year]	1.0946418	0.1145239	91.35908	<.0001*
Vehicle_Age[> 2 Years]	-0.686601	0.1129562	36.947778	<.0001*
Vehicle_Damage[No]	1.2089982	0.1626766	55.233323	<.0001*
Previously_Insured	3.3839828	0.6109719	30.677068	<.0001*

Model Accuracy

Metric	Value	Interpretation
Whole Model Chi-Square	1152.334	Very strong evidence model improves over null model
Whole Model p-value	< .0001	Model is highly significant
Pearson Goodness-of-Fit Chi-Square	4271.242	No evidence of lack of fit
Pearson p-value	1.0000	Excellent model fit
Deviance Goodness-of-Fit Chi-Square	3152.115	No evidence of lack of fit
Deviance p-value	1.0000	Excellent model fit
AICc	3164.129	Good relative model quality



Fitting Response vs Predicted
Response yields:

- AUC = 0.084185 → Excellent discrimination between responses
- pseudo- R^2 = 0.2337 → good predictive model
- ROC is far above the diagonal → good model performance

Conclusion: Model is a good predictor of future responses

Model Validation

Assumptions of Logistic Regression:

1. Dependent variable is binary – Yes (Response = Yes/No)
2. Independence of observations – Yes (Each record is a unique customer lead)
3. Large Sample Size – Yes
4. No extreme outliers – Less than 3% of the Pearson residuals are >2 or <-2.
5. No perfect multicollinearity (vs ID as a stand-in continuous variable)

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	3377.593	155.1601	21.77	<.0001*	.
Gender[Female]	-33.67542	26.90704	-1.25	0.2108	1.0355933
Age	2.6230636	2.888018	0.91	0.3638	2.9173453
Vehicle_Age[< 1 Year]	86.217896	77.38576	1.11	0.2653	8.237015
Vehicle_Age[> 2 Years]	-207.8423	95.78526	-2.17	0.0301*	4.4224797
Vehicle_Damage[No]	32.146309	48.47222	0.66	0.5072	3.3843676
Annual Premium	-0.179594	0.224026	-0.80	0.4228	1.0533923
Previously_Insured	-42.96716	96.72585	-0.44	0.6569	3.3466165

While one of the vehicle dummy variable shows a VIF over 5, it is not over 10 and is expected to due to close relation to the other dummy variables. The data shows no perfect multicollinearity.

6. The Box-Tidwell Linearity Test for the continuous variable (Age and Annual Premium) confirms linearity.

Parameter Estimates						
Term	Estimate	Std Error	ChiSquare	Prob>ChiSq	Lower CL	Upper CL
Intercept	4.5189417	1.2470073	13.13214	0.0003*	2.0743456	6.9635378
Gender[Female]	0.0514737	0.0458366	1.2610885	0.2614	-0.038383	0.1413304
Age	-0.481948	0.1199177	16.152266	<.0001*	-0.717031	-0.246865
Vehicle_Age[< 1 Year]	0.6778422	0.1503704	20.32042	<.0001*	0.3830606	0.9726239
Vehicle_Age[> 2 Years]	-0.491476	0.1238917	15.736921	<.0001*	-0.734349	-0.248602
Vehicle_Damage[No]	1.2058428	0.16296	54.754424	<.0001*	0.8863808	1.5253048
Annual Premium	0.0244819	0.0096805	6.3957371	0.0114*	0.0055044	0.0434593
Previously_Insured	3.4062621	0.6120055	30.977438	<.0001*	2.2065047	4.6060196
Age_LogInteract	0.1039276	0.0245991	17.849389	<.0001*	0.0557043	0.152151
Premium_LogInteract	-0.003426	0.0013556	6.3874483	0.0115*	-0.006083	-0.000769

Conclusion: Assumptions for Logistic Regression are met.

Cross Validation

Metric	Training Set	Test Set	Interpretation
AUC	0.84437	0.83684	Very close → model generalizes well
Misclassification Rate	0.1209	0.1286	Only slightly worse on test set (good)
Entropy R^2	0.2362	0.2282	stable across sets
Generalized R^2	0.3054	0.2989	Very similar → no overfitting
Mean –Log p	0.2775	0.2927	Slight increase on test set (normal)
RASE	0.2958	0.3054	Minor increase → acceptable
N	4,137	1,711	70/30 split confirmed

Sample of Recommended List of Leads to Contact

See “Top Conversions.docx” for full list

ID	GENDER	AGE	VEHICLE AGE	VEHICLE DAMAGE	ANNUAL PREMIUM	VINTAGE	PREVIOUSLY INSURED	PREDICTED RESPONSE
82939	Male	34	< 1 Year	No	387.28	71	1	0.999622
80795	Female	33	< 1 Year	No	403.37	211	1	0.999612
84290	Male	33	< 1 Year	No	291.94	109	1	0.999612
85369	Male	33	< 1 Year	No	303.96	118	1	0.999612
78929	Female	32	< 1 Year	No	255.77	262	1	0.999602
80842	Female	32	< 1 Year	No	233.72	64	1	0.999602
81170	Female	32	< 1 Year	No	254.43	236	1	0.999602
83891	Female	32	< 1 Year	No	309.2	62	1	0.999602
83964	Female	32	< 1 Year	No	410.21	108	1	0.999602
84171	Female	32	< 1 Year	No	367.93	284	1	0.999602
84806	Female	32	< 1 Year	No	361.89	72	1	0.999602
85331	Female	32	< 1 Year	No	259.63	121	1	0.999602
80117	Female	31	< 1 Year	No	262.66	167	1	0.999592
80521	Female	31	< 1 Year	No	490.99	172	1	0.999592
80698	Male	31	< 1 Year	No	297.82	40	1	0.999592
80732	Male	31	< 1 Year	No	248.15	46	1	0.999592
80738	Female	31	< 1 Year	No	274.58	65	1	0.999592
81148	Male	31	< 1 Year	No	222.67	217	1	0.999592
82695	Male	31	< 1 Year	No	351.52	84	1	0.999592
83315	Female	31	< 1 Year	No	279.53	268	1	0.999592
84628	Female	31	< 1 Year	No	329.42	209	1	0.999592
84700	Female	31	< 1 Year	No	240.5	261	1	0.999592

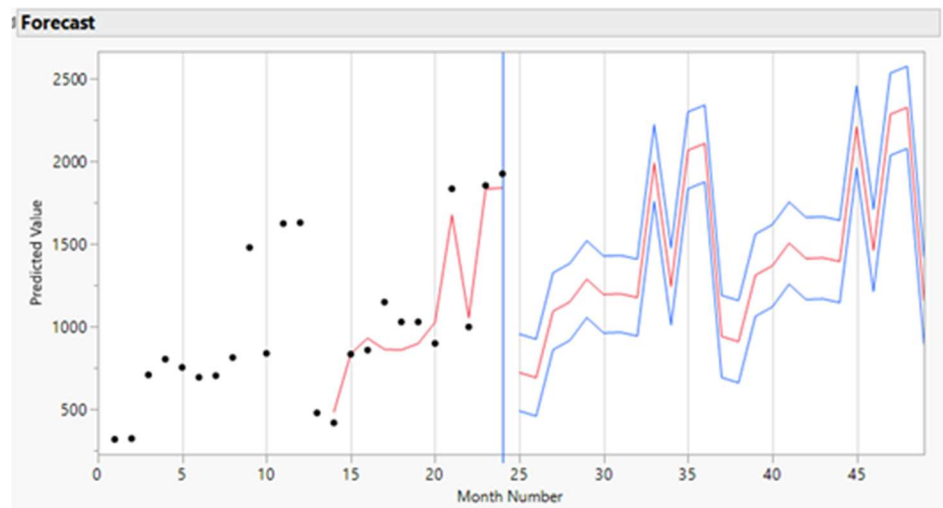
Appendix 4: Time Series Sales Forecast with Seasonality

Car sales in expansion regions show upward trend with noticeable seasonality, especially at the end of the calendar year. Comparing several exponential smoothing models show that the Winter-Holts method is best for this instance because it captures both seasonality and overall trends.

Comparison of Exponential Smoothing Models (JMP Output Summary)

CRITERIA	SIMPLE	SEASONAL	HOLT-WINTERS ADDITIVE
TREND COMPONENT?	✗ No	✗ No	✓ Yes
SEASONAL COMPONENT?	✗ No	✓ Yes	✓ Yes
RSQUARE	0.111 (very poor)	0.922 (excellent)	0.921 (excellent)
RSQUARE ADJ	—	0.913	0.903
MAPE	35.6 (poor accuracy)	10.25 (excellent)	10.23 (excellent)
MAE	325.1 (poor accuracy)	105.1 (excellent)	106.5 (excellent)
AIC	346.79 (worst)	142.76 (best)	144.87 (very good)
SBC / BIC	347.92	143.56	146.07
MODEL STABLE?	✓ Yes	✓ Yes	✓ Yes
FITS DATA TREND?	✗ No	✗ No	✓ Yes
FITS SEASONALITY?	✗ No	✓ Yes	✓ Yes
FORECAST SUITABILITY	✗ Very poor (ignores pattern)	⚠ Good but under-forecasts (ignores trend)	★ Best choice (captures trend + seasonality)

2026 Monthly Car Sales Forecast



MONTH	2025 ACTUAL(1000S)	2026 FORECAST(1000S)	% TOTAL SALES
JAN	475	719.02	5%
FEB	415	688.17	4%
MAR	830	1089.72	7%
APR	855	1146.12	7%
MAY	1145	1283.61	8%
JUN	1025	1190.52	7%
JUL	1025	1195	8%
AUG	895	1172.66	7%
SEP	1830	1986.55	13%
OCT	995	1239.02	8%
NOV	1850	2062.62	13%
DEC	1921	2104.01	13%

While year over year increase is projected to be nearly 20%, sales are much higher in the last 4 months of the year which accounts for over 47% of sales. Assuming that ABC Insurance captures an equitable portion of market share in the expansion territories, for staffing considerations, an increase of 20% service levels is required based on increasing sales over the year. We recommend full staffing levels by mid-year to accommodate the seasonal spike in the fall months.

Appendix 5: Data Sources

Data Sources:

<https://www.kaggle.com/datasets/anmolkumar/health-insurance-cross-sell-prediction>

<https://www.kaggle.com/datasets/missionjee/car-sales-report/data>